

SECTION 01

Design and Simulation of Half Adder & Full Adder

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1. Introduction

An adder is an essential combinational logic circuit specifically designed to execute binary addition operations across computational units. In modern digital computer architecture, adders form the structural core of the Arithmetic Logic Unit (ALU) to process arithmetic calculations and instruction execution.

Depending on operational requirements and input data width, binary adders are primarily categorized into Half Adders and Full Adders, which handle two-bit and three-bit additions respectively. Graphical software tools like Logisim provide an intuitive digital workbench to model, simulate, and visually verify the exact logic flow and functional timing of these foundational logic gates.

Key Takeaway: Adders represent the core arithmetic building blocks of ALUs. Simulation in Logisim ensures accurate logic mapping prior to hardware deployment.

SECTION 02

2. Half Adder Circuit

A Half Adder is a basic digital building block engineered to compute the arithmetic sum of two single-bit binary inputs, designated as input **A** and input **B**. The circuit generates two discrete binary outputs: the Sum (**S**) representing the least significant output bit, and the Carry (**C**) representing the high-order bit overflow.

Internally, an XOR gate determines the Sum logic, while an AND gate generates the Carry output whenever both input bits are simultaneously high.

BOOLEAN LOGIC EXPRESSIONS

$$\text{Sum (S)} = A \oplus B$$

$$\text{Carry (C)} = A \cdot B$$

Although highly efficient for processing basic single-bit operational tasks, the Half Adder cannot accept an external carry bit from a preceding lower-significant stage, restricting its standalone application in multi-bit arithmetic calculations.

SECTION 03

3. Full Adder Circuit

A Full Adder expands computational capabilities by adding three simultaneous binary inputs: two primary operand bits (**A** and **B**) alongside an incoming carry bit (C_{in}) from a lower-significant arithmetic stage. The circuit computes two principal binary outputs: the final arithmetic Sum (**S**) and an outgoing Carry-out (C_{out}).

A Full Adder is modularly constructed by interconnecting two individual Half Adders along with a single OR gate to consolidate carry signals.

BOOLEAN LOGIC EXPRESSIONS

$$\text{Sum (S)} = A \oplus B \oplus C_{in}$$

$$\text{Carry-out (C}_{out}\text{)} = (A \cdot B) + C_{in}(A \oplus B)$$

By supporting incoming carry signals, Full Adders enable cascaded multi-bit adder configurations (such as ripple-carry adders), making them crucial components in CPU computational hardware.

SECTION 04

4. Logisim Simulation & Conclusion

The schematic implementation and signal verification inside Logisim follows a structured step-by-step methodology:

1. **Environment Setup:** Open Logisim and configure the schematic grid layout workspace.
2. **Gate Placement:** Instantiate XOR, AND, and OR logic gate modules onto the workspace canvas.
3. **Wiring & I/O Pins:** Wire input pin toggles (A, B, C_{in}) to gate terminals and attach output pin probes (Sum, Carry).
4. **Logic Testing:** Systematically toggle input switches through all binary permutations.
5. **Truth Table Verification:** Inspect LED output indicators to verify exact match with theoretical truth table outputs.

Conclusion: Half and Full Adders form the essential foundation of modern binary logic arithmetic. Simulation software like Logisim streamlines circuit design, visual timing evaluation, and debugging, facilitating the creation of complex multi-bit adders and advanced ALU components.